

1).

**PROGRAM:**

TITLE KEYSTATE.ASM -- Program to display keyboard special keys state

```
;/*****  
*****
```

```
;//* *
```

```
;//* Copyright (c) 1985-2022, American Megatrends International LLC. *
```

```
;//* *
```

```
;//* All rights reserved. Subject to AMI licensing agreement. *
```

```
;//* *
```

```
;/*****  
*****
```

```
;<AMI_FHDR_START>
```

```
;
```

```
; Name: <keystate.asm>
```

```
;
```

```
; Description:
```

```
; <Program to display keyboard special keys state>
```

```
;
```

```
;<AMI_FHDR_END>
```

```
.******  
,*  
*****
```

```
.model small
```

```
.stack
```

```
;-----
```

```
data segment
```

```
MSG1 DB 13,10,'Keyboard Special Keys State',13,10,'$'
```

```
MSG2 DB 'Scroll Lock : $'
MSG3 DB 13,10,'Num Lock  : $'
MSG4 DB 13,10,'Caps Lock  : $'
MSG5 DB 13,10,'Insert    : $'
ONM  DB 'ON$'
OFFM DB 'OFF$'
```

```
data ends
```

```
;-----
```

```
code segment
```

```
assume cs:code, ds:data
```

```
start:
```

```
    MOV AX, data
```

```
    MOV DS, AX
```

```
    LEA DX, MSG1
```

```
    MOV AH, 09H
```

```
    INT 21H
```

```
    MOV AH, 02H
```

```
    INT 16H          ; AL = keyboard status flags
```

```
    MOV BL, AL
```

```
    LEA DX, MSG2
```

```
    MOV AH, 09H
```

```
    INT 21H
```

```
    TEST BL, 10H
```

```
    JZ SCROLL_OFF
```

```
    LEA DX, ONM
```

```
    JMP SCROLL_SHOW
```

SCROLL\_OFF:

LEA DX, OFFM

SCROLL\_SHOW:

MOV AH, 09H

INT 21H

LEA DX, MSG3

MOV AH, 09H

INT 21H

TEST BL, 20H

JZ NUM\_OFF

LEA DX, ONM

JMP NUM\_SHOW

NUM\_OFF:

LEA DX, OFFM

NUM\_SHOW:

MOV AH, 09H

INT 21H

LEA DX, MSG4

MOV AH, 09H

INT 21H

TEST BL, 40H

JZ CAPS\_OFF

LEA DX, ONM

JMP CAPS\_SHOW

CAPS\_OFF:

LEA DX, OFFM

CAPS\_SHOW:

```
MOV AH, 09H
INT 21H
LEA DX, MSG5
MOV AH, 09H
INT 21H
TEST BL, 80H
JZ INS_OFF
LEA DX, ONM
JMP INS_SHOW
```

INS\_OFF:

```
LEA DX, OFFM
```

INS\_SHOW:

```
MOV AH, 09H
INT 21H
MOV AH, 4CH
INT 21H
```

code ends

end start

;-----

**OUTPUT:**

```
C:\>keystate.asm
Illegal command: keystate.asm.

C:\>keystate

Keyboard Special Keys State
Scroll Lock : OFF
Num Lock    : OFF
Caps Lock   : OFF
Insert      : OFF
C:\>
```

2)

**PROGRAM:**

TITLE EBDAINFO.ASM -- Program to display EBDA start address and size

```
;/*****  
*****
```

```
;//* *
```

```
;//* Copyright (c) 1985-2022, American Megatrends International LLC. *
```

```
;//* *
```

```
;//* All rights reserved. Subject to AMI licensing agreement. *
```

```
;//* *
```

```
;/*****  
*****
```

```
; <AMI_FHDR_START>
```

```
;
```

```
; Name: <EBDA.asm>
```

```
;
```

```
; Description:
```

```
; <Program to display the starting address and size of EBDA>
```

```
;
```

```
; <AMI_FHDR_END>
```

```
.******  
,*  
*****
```

```
.model small
```

```
.stack
```

```
;-----
```

```
data segment
```

```
MSG1 DB 13,10,'EBDA Information',13,10,'$'
```

MSG2 DB 'EBDA Start Segment : \$'

MSG3 DB 13,10,'EBDA Size (KB) : \$'

EBDSEG DW 0000H

EBDSIZ DW 0000H

data ends

;-----

code segment

assume cs:code, ds:data

start:

MOV AX, data

MOV DS, AX

LEA DX, MSG1

MOV AH, 09H

INT 21H

; read EBDA segment from BDA 40:0E

MOV AX, 40H

MOV ES, AX

MOV AX, ES:[0EH]

MOV EBDSEG, AX

LEA DX, MSG2

MOV AH, 09H

INT 21H

MOV AX, EBDSEG

CALL PRINT\_HEX16

; conventional memory size in KB

INT 12H ; AX = KB before EBDA / top conventional

```

; EBDA size = AX - (EBDA segment / 64)
MOV BX, EBDSEG
MOV CL, 6
SHR BX, CL      ; BX = segment / 64 = KB base
SUB AX, BX
MOV EBDSIZ, AX
LEA DX, MSG3
MOV AH, 09H
INT 21H
MOV AX, EBDSIZ
CALL PRINT_HEX16
MOV AH, 4CH
INT 21H

```

```

;-----

```

```

; PRINT_HEX16

```

```

;-----

```

```

PRINT_HEX16 PROC

```

```

    PUSH AX

```

```

    PUSH BX

```

```

    PUSH CX

```

```

    PUSH DX

```

```

    MOV BX, AX

```

```

    MOV CX, 4

```

```

P1:

```

```

    ROL BX, 1

```

```

    ROL BX, 1

```

```
ROL BX, 1
ROL BX, 1
MOV DL, BL
AND DL, 0FH
CMP DL, 09H
JBE P2
ADD DL, 07H
```

P2:

```
ADD DL, 30H
MOV AH, 02H
INT 21H
LOOP P1
POP DX
POP CX
POP BX
POP AX
RET
```

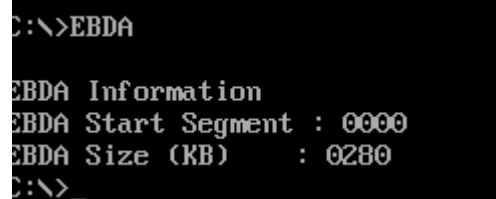
PRINT\_HEX16 ENDP

code ends

end start

;-----

## OUTPUT:



```
C:\>EBDA
EBDA Information
EBDA Start Segment : 0000
EBDA Size (KB)      : 0280
C:\>
```

3)

**PROGRAM:**

TITLE OPROM.ASM -- Program to verify Option ROMs and call VGA  
OpROM

```
;*****  
*****
```

```
;/* *
```

```
;/* Copyright (c) 1985-2022, American Megatrends International LLC. *
```

```
;/* *
```

```
;/* All rights reserved. Subject to AMI licensing agreement. *
```

```
;/* *
```

```
;*****  
*****
```

```
;<AMI_FHDR_START>
```

```
;
```

```
; Name: <ROM.asm>
```

```
;
```

```
; Description:
```

```
; <Program to verify Option ROM presence, count them, and transfer control to  
VGA OpROM>
```

```
;<AMI_FHDR_END>
```

```
.*****  
;
```

```
.model small
```

```
.stack
```

```
;-----
```

```
data segment
```

MSG1 DB 13,10,'Option ROM Scan',13,10,'\$'

MSG2 DB 'Option ROM Count : \$'

MSG3 DB 13,10,'VGA Option ROM found. Transferring control...\$'

MSG4 DB 13,10,'Control returned from VGA Option ROM.\$'

MSG5 DB 13,10,'VGA Option ROM not found.\$'

COUNT DB 00H

VGAPTR DW 0003H,0C000H

data ends

;-----

code segment

assume cs:code, ds:data

start:

MOV AX, data

MOV DS, AX

LEA DX, MSG1

MOV AH, 09H

INT 21H

MOV COUNT, 00H

MOV AX, 0C000H

SCAN\_LOOP:

CMP AX, 0E000H

JAE SHOW\_COUNT

MOV ES, AX

MOV BX, ES:[0000H]

CMP BX, 0AA55H

JNE NEXT\_SEG

INC COUNT

NEXT\_SEG:

ADD AX, 0080H

JMP SCAN\_LOOP

SHOW\_COUNT:

LEA DX, MSG2

MOV AH, 09H

INT 21H

MOV AL, COUNT

CALL PRINT\_HEX8

; check VGA OpROM at C000:0000

MOV AX, 0C000H

MOV ES, AX

MOV BX, ES:[0000H]

CMP BX, 0AA55H

JNE NO\_VGA

LEA DX, MSG3

MOV AH, 09H

INT 21H

CALL DWORD PTR VGAPTR

LEA DX, MSG4

MOV AH, 09H

INT 21H

JMP EXIT1

NO\_VGA:

LEA DX, MSG5

MOV AH, 09H

INT 21H

EXIT1:

MOV AH, 4CH

INT 21H

;-----

; PRINT\_HEX8

;-----

PRINT\_HEX8 PROC

PUSH AX

PUSH BX

PUSH DX

MOV BL, AL

MOV DL, BL

SHR DL, 1

SHR DL, 1

SHR DL, 1

SHR DL, 1

AND DL, 0FH

CMP DL, 09H

JBE D1

ADD DL, 07H

D1:

ADD DL, 30H

MOV AH, 02H

INT 21H

MOV DL, BL

AND DL, 0FH

CMP DL, 09H

JBE D2

ADD DL, 07H

D2:

ADD DL, 30H

MOV AH, 02H

INT 21H

POP DX

POP BX

POP AX

RET

PRINT\_HEX8 ENDP

code ends

end start

;-----

## OUTPUT:



```
C:\>ROM
Option ROM Scan
Option ROM Count : 01
VGA Option ROM found. Transferring control..._
```